

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

593000602 - Mobile Communication Systems

DEGREE PROGRAMME

59AJ - Master Universitario En Comunicaciones Inalámbricas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	593000602 - Mobile Communication Systems
No of credits	6 ECTS
Type	Compulsory
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	59AJ - Master Universitario en Comunicaciones Inalámbricas
Centre	59 - E.T.S. De Ingeniería Y Sist. De Telecom.
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Juan Anton Moreno Garcia-Loygorri	8418	juan.moreno.garcia-loygorri@upm.es	Sin horario.
Antonio Perez Yuste (Subject coordinator)	8304	antonio.perez@upm.es	Sin horario.
Angel Martinez Jimenez	7010	angel.martinez.jimenez@upm.es	Sin horario.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

2.2. Research assistants

Name and surname	Email	Faculty member in charge
Yao ., Xiao	xiao.yao@alumnos.upm.es	Perez Yuste, Antonio

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

- Network Architecture And Protocols

3.2. Other recommended learning outcomes

- Search of technical information: databases, standards, papers, etc.
- Use of RF software: MATLAB
- Operation of RF instrumentation
- Telecommunication Systems

4. Skills and learning outcomes *

4.1. Skills to be learned

CB6 - Poseer y comprender conocimientos que aporten una base u oportunidad de ser originales en el desarrollo y/o aplicación de ideas, a menudo en un contexto de investigación/ Students have demonstrated knowledge and understanding providing the groundwork or opportunity for innovation in developing and/or applying ideas, often within a research context

CB7 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio/Students are capable of applying their knowledge, understanding, and problem-solving abilities in new or unfamiliar environments within broader (or multidisciplinary) contexts related to their field of study

CB8 - Que los estudiantes sean capaces de integrar conocimientos y enfrentarse a la complejidad de formular juicios a partir de una información que, siendo incompleta o limitada, incluya reflexiones sobre las responsabilidades sociales y éticas vinculadas a la aplicación de sus conocimientos y juicios/Students are capable

of integrating knowledge and making complex decisions, which, although based on incomplete or limited information, require reflection on social and ethical responsibilities linked to the application of their knowledge and opinions

CEM03 - Analizar y diseñar la arquitectura, servicios y protocolos de la red de acceso radio de un sistema de comunicaciones móviles 4G y 5G / Analyze and design the architecture, services, and protocols of the radio access network in a 4G or 5G mobile communication system.

CGI02 - Comprender el procedimiento, valor y límites del método científico, siendo capaz de identificar, localizar y obtener datos requeridos en un trabajo de investigación, de diseñar y guiar investigaciones analíticas, de modelado y experimentales, así como de evaluar datos de una manera crítica y extraer conclusiones. / Understand the procedure, value, and limits of the scientific method, being able to identify, locate and obtain data required in a research work, to design and guide analytical, modeling, and experimental investigations, as well as to critically evaluate data and extract conclusions.

CGI03 - Valorar la importancia de las fuentes documentales, manejarlas y buscar la información para el desarrollo de cualquier trabajo de investigación. / Assess the importance of documentary sources, manage them and search for information for the development of any research work.

CGI04 - Leer y comprender publicaciones dentro de su ámbito de estudio/investigación, así como su catalogación y valor científico. / Read and understand publications within their field of study / research, as well as their cataloging and scientific value.

UPM1 - Uso de la lengua inglesa / Use of the English language

UPM4 - Organización y planificación / Planning and organization

UPM5 - Gestión de la Información / Information management

4.2. Learning outcomes

RA7 - Aplicar el conocimiento adquirido a la solución de problemas cualitativos y cuantitativos relacionados con las comunicaciones móviles personales/ Apply the knowledge acquired to the solution of qualitative and quantitative problems related to personal mobile communications

RA8 - Identificar el impacto social y económico de las comunicaciones móviles en un contexto global/ Identify the social and economic impact of mobile communications in a global context.

RA3 - Elegir los métodos y herramientas matemáticas necesarios para abordar la solución de un problema/ Choose the mathematical methods and tools necessary to tackle the solution of a problem

RA2 - Aplicar el conocimiento adquirido a la comprensión de problemas cualitativos y cuantitativos relacionados con la codificación fuente y la codificación de canal/ Apply the knowledge acquired to the understanding of qualitative and quantitative problems related to source coding and channel coding

RA15 - Diseñar y desarrollar los componentes de una red de comunicaciones móviles y sus protocolos a partir de especificaciones técnicas/ Design and develop the components of a mobile communications network and its protocols based on technical specifications

RA5 - Interpretar los datos derivados de observaciones empíricas y mediciones en términos de su importancia y relacionarlos con la teoría apropiada/ Interpret data derived from empirical observations and measurements in terms of their importance and relate them to the appropriate theory

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Mobile communications have evolved into a remarkable global service, achieving near-universal coverage with penetration rates close to 100% today. Socially, they have become an indispensable part of modern life, while economically, they serve as a critical enabler for businesses worldwide. LTE (Long-Term Evolution), the fourth generation (4G) of mobile technology, remains the most widely adopted wireless broadband standard globally. Although LTE continues to advance and will remain in use for years to come, the introduction of 5G New Radio (NR) marks the next step in meeting future connectivity demands.

While enhanced mobile broadband remains a primary goal, future wireless networks will also support a broader range of innovative applications. These include mission-critical communications, smart energy grids, autonomous vehicles and intelligent transportation systems, remote healthcare and telesurgery, and expanded industrial IoT, among others.

This course provides a clear and comprehensive understanding of current and next-generation mobile communication systems?particularly 4G-LTE and 5G-NR?with a focus on the Radio Access Network (RAN) and the Physical Layer (PHY). We will examine both technologies in parallel, exploring foundational concepts as well as the latest advancements.

5.2. Syllabus

1. LTE and 5G overview
 - 1.1. Public Land Mobile Networks (PLMN) and NG-RAN
 - 1.2. Stakeholder Roles in a Modern PLMN
 - 1.3. RAN Technology Enablers
 - 1.4. 5G Radio Scenarios
 - 1.5. 5G Rollout in the World
 - 1.6. 5G End-to-end KPIs
 - 1.7. 6G Expected Features
2. Architecture of the Radio Access Network
 - 2.1. E-UTRAN Architecture
 - 2.2. 5G-NR Architecture
 - 2.3. Multi-Radio Dual Connectivity
 - 2.4. Radio Cell Features
 - 2.5. Radio Network IDs
 - 2.6. RRC State Diagram
 - 2.7. Quality of Service
 - 2.8. Air Interface Protocol Stack and Channels
 - 2.9. The Mobile Spectrum
3. PHY transmission structure
 - 3.1. Duplexing scheme
 - 3.2. Frequency-domain structure
 - 3.3. Bandwidth parts
 - 3.4. Time-domain structure
 - 3.5. Synchronization signal block (SSB)
 - 3.6. Antenna ports
 - 3.7. Resource grid
4. PHY signal processing: modulation and channel coding

- 4.1. General scheme of a communication system
- 4.2. The wireless channel
- 4.3. Modulation and demodulation
- 4.4. Error management
- 4.5. Adaptive modulation and coding
- 4.6. OFDM basics
- 4.7. The cyclic prefix
- 4.8. OFDMA basics
- 4.9. Channel coding
- 5. PHY signal processing: multi-antenna transmission
 - 5.1. Beamforming
 - 5.2. Spatial multiplexing
 - 5.3. Massive MIMO
 - 5.4. Downlink and uplink precoding
 - 5.5. Beam management
- 6. Air Interface L1
 - 6.1. Layer 2 summary
 - 6.2. Layer 1 review
 - 6.3. Synchronisation signal block
 - 6.4. Control resource set
 - 6.5. Channel state information
 - 6.6. Data transmission
- 7. Cell Acquisition
 - 7.1. Acquisition procedure
 - 7.2. Resource mapping
 - 7.3. Acquisition of the SS/PBCH block
 - 7.4. System information
- 8. Link Adaptation
 - 8.1. CSI reference signals

8.2. Channel state information

8.3. Physical Uplink Control Channel

9. Mm-wave for wireless communications

9.1. Spectrum for NR flexibility

9.2. Radio wave propagation for Mm-wave

9.3. Mm-wave standardization

9.4. Types of base stations. Requirements.

9.5. Antennas and arrays for mm-Wave applications

10. Channel sounding

10.1. Motivation

10.2. Channel modelling

10.3. Propagation measurements in extreme conditions

10.4. Channel models in vehicular scenarios

11. AI-powered radio access networks

11.1. ML-Modeling with ORAN

11.2. Non-Real-Time (Non-RT) RIC

11.3. Near-Real-Time (Near-RT) RIC

11.4. O-DU and O-RU AI control loop

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	T1. LTE and 5G overview Duration: 02:00 T2. Architecture of the Radio Access Network Duration: 04:00	T2. Architecture of the Radio Access Network Duration: 02:00		T1-T2. LTE and 5G overview (Quizz) Progressive assessment and Global Examination Not Presential Duration: 01:00 T2. Architecture of the Radio Access Network (Workshop) Progressive assessment and Global Examination Presential Duration: 02:00
2	T3. PHY transmission structure Duration: 04:00	T3. PHY transmission structure Duration: 04:00		
3	T4. PHY signal processing: modulation and channel coding Duration: 04:00	T3. PHY transmission structure Duration: 04:00		T3-T4. PHY Transmission Structure and Modulation (Quizz) Progressive assessment and Global Examination Not Presential Duration: 01:00 T3. PHY transmission structure (Workshop) Progressive assessment and Global Examination Presential Duration: 02:00
4	T5. PHY signal processing: multi-antenna transmission Duration: 06:00	T4. PHY signal processing: modulation and channel coding Duration: 02:00		T4. PHY signal processing: modulation and channel coding (Workshop) Progressive assessment and Global Examination Presential Duration: 02:00 T5. PHY signal procession: multi antenna transmission (Quizz) Progressive assessment and Global Examination Not Presential Duration: 01:00

5	T6. Air Interface L1 Duration: 04:00 T7. Cell Acquisition Duration: 01:00 T8. Link Adaptation Duration: 01:00	T5. PHY signal processing: multi-antenna transmission Duration: 02:00		T5. PHY signal processing: multi-antenna transmission (Workshop) Progressive assessment and Global Examination Presential Duration: 02:00 T7-T8. Cell Acquisition and Link Adaptation (Quizz) Progressive assessment and Global Examination Not Presential Duration: 01:00
6	T9. Mm-wave for wireless communications Duration: 06:00 T10. Channel sounding Duration: 02:00			T9. Mm-wave for wireless communications (Quizz) Progressive assessment and Global Examination Not Presential Duration: 01:00
7	T10. Channel sounding Duration: 02:00 T11. AI-powered radio access networks Duration: 04:00	T10. Channel sounding Duration: 02:00		T10. Channel Sounding (Workshop) Progressive assessment and Global Examination Presential Duration: 02:00
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Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
1	T1-T2. LTE and 5G overview (Quizz)		No Presential	01:00	10%	5 / 10	UPM1 UPM4 UPM5 CEM03
1	T2. Architecture of the Radio Access Network (Workshop)		Face-to-face	02:00	10%	5 / 10	CB7 CGI03 UPM1 UPM5 CEM03
3	T3-T4. PHY Transmission Structure and Modulation (Quizz)		No Presential	01:00	15%	5 / 10	CB8 CGI04 UPM1 CEM03
3	T3. PHY transmission structure (Workshop)		Face-to-face	02:00	10%	5 / 10	CB7 CGI03 UPM1 UPM5 CEM03
4	T4. PHY signal processing: modulation and channel coding (Workshop)		Face-to-face	02:00	10%	5 / 10	CB6 CB7 CGI02 CGI03 UPM1 UPM5 CEM03
4	T5. PHY signal procession: multi antenna transmission (Quizz)		No Presential	01:00	10%	5 / 10	CB7 CGI02 CGI03 CGI04 UPM5
5	T5. PHY signal processing: multi-antenna transmission (Workshop)		Face-to-face	02:00	10%	5 / 10	CB8 CGI03 UPM1 UPM4 CEM03

5	T7-T8. Cell Acquisition and Link Adaptation (Quizz)		No Presential	01:00	5%	5 / 10	CB8 CGI02 UPM1 UPM5
6	T9. Mm-wave for wireless communications (Quizz)		No Presential	01:00	10%	5 / 10	CB6 CGI03 CGI04 UPM1 UPM4 UPM5
7	T10. Channel Sounding (Workshop)		Face-to-face	02:00	10%	5 / 10	CGI03 CGI04 UPM1 UPM4 UPM5 CEM03

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
1	T1-T2. LTE and 5G overview (Quizz)		No Presential	01:00	10%	5 / 10	UPM1 UPM4 UPM5 CEM03
1	T2. Architecture of the Radio Access Network (Workshop)		Face-to-face	02:00	10%	5 / 10	CB7 CGI03 UPM1 UPM5 CEM03
3	T3-T4. PHY Transmission Structure and Modulation (Quizz)		No Presential	01:00	15%	5 / 10	CB8 CGI04 UPM1 CEM03
3	T3. PHY transmission structure (Workshop)		Face-to-face	02:00	10%	5 / 10	CB7 CGI03 UPM1 UPM5 CEM03
4	T4. PHY signal processing: modulation and channel coding (Workshop)		Face-to-face	02:00	10%	5 / 10	CB6 CB7 CGI02 CGI03 UPM1 UPM5 CEM03
4	T5. PHY signal procession: multi antenna transmission (Quizz)		No Presential	01:00	10%	5 / 10	CB7 CGI02 CGI03 CGI04 UPM5

5	T5. PHY signal processing: multi-antenna transmission (Workshop)		Face-to-face	02:00	10%	5 / 10	CB8 CGI03 UPM1 UPM4 CEM03
5	T7-T8. Cell Acquisition and Link Adaptation (Quizz)		No Presential	01:00	5%	5 / 10	CB8 CGI02 UPM1 UPM5
6	T9. Mm-wave for wireless communications (Quizz)		No Presential	01:00	10%	5 / 10	CB6 CGI03 CGI04 UPM1 UPM4 UPM5
7	T10. Channel Sounding (Workshop)		Face-to-face	02:00	10%	5 / 10	CGI03 CGI04 UPM1 UPM4 UPM5 CEM03

7.1.3. Referred (re-sit) examination

No se ha definido la evaluación extraordinaria.

7.2. Assessment criteria

The type of course and the learning approach described in this guide make more convenient to select an assessment mechanism different to the traditional final exam. A progressive evaluation methodology is here proposed for this course, based on a set of short quizzes and practical workshops. This way, main concepts can be properly set up while the attractiveness of contents are increased. The assessment of workshop activities are based on the realisation of a report by students. This report will be also evaluated in order to get the final grade.

Those students who do not meet the goals and do not pass the regular activities, will be able to attend an extraordinary exam organised on July. This will consist of two parts: firstly, a questionnaire selected among all topics in this course must be responded on-line and, secondly, a lab practice with several exercises related to the practical activities carried out along the course must be duly done. It is mandatory to pass the first part before moving to the second part, which also needs to be passed in order to get the final grade.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Slide shows	Web resource	Moodle LMS
Lab scripts	Web resource	Moodle LMS
3GPP	Web resource	The 3rd Generation Partnership Project, http://www.3gpp.org/
The 3G4G Blog	Web resource	https://blog.3g4g.co.uk
Cox, 2021	Bibliography	Cox, C., "An Introduction to 5G. The New Radio, 5G Network and Beyond", Ed. John Wiley & Sons, 2021
Cox, 2014	Bibliography	Cox, C., "An Introduction to LTE. LTE-Advanced, SAE, VoLTE and 4G Mobile Communications", 2nd ed, Ed. John Wiley & Sons, 2014
Dahlman, 2018	Bibliography	Dahlman, E. et al, "5G NR: The Next Generation Wireless Access Technology", Academic Press, Elsevier, 2018
Dahlman, 2011	Bibliography	Dahlman, E. et al, "4G: LTE/LTE-Advanced for Mobile Broadband", Academic Press, 2011